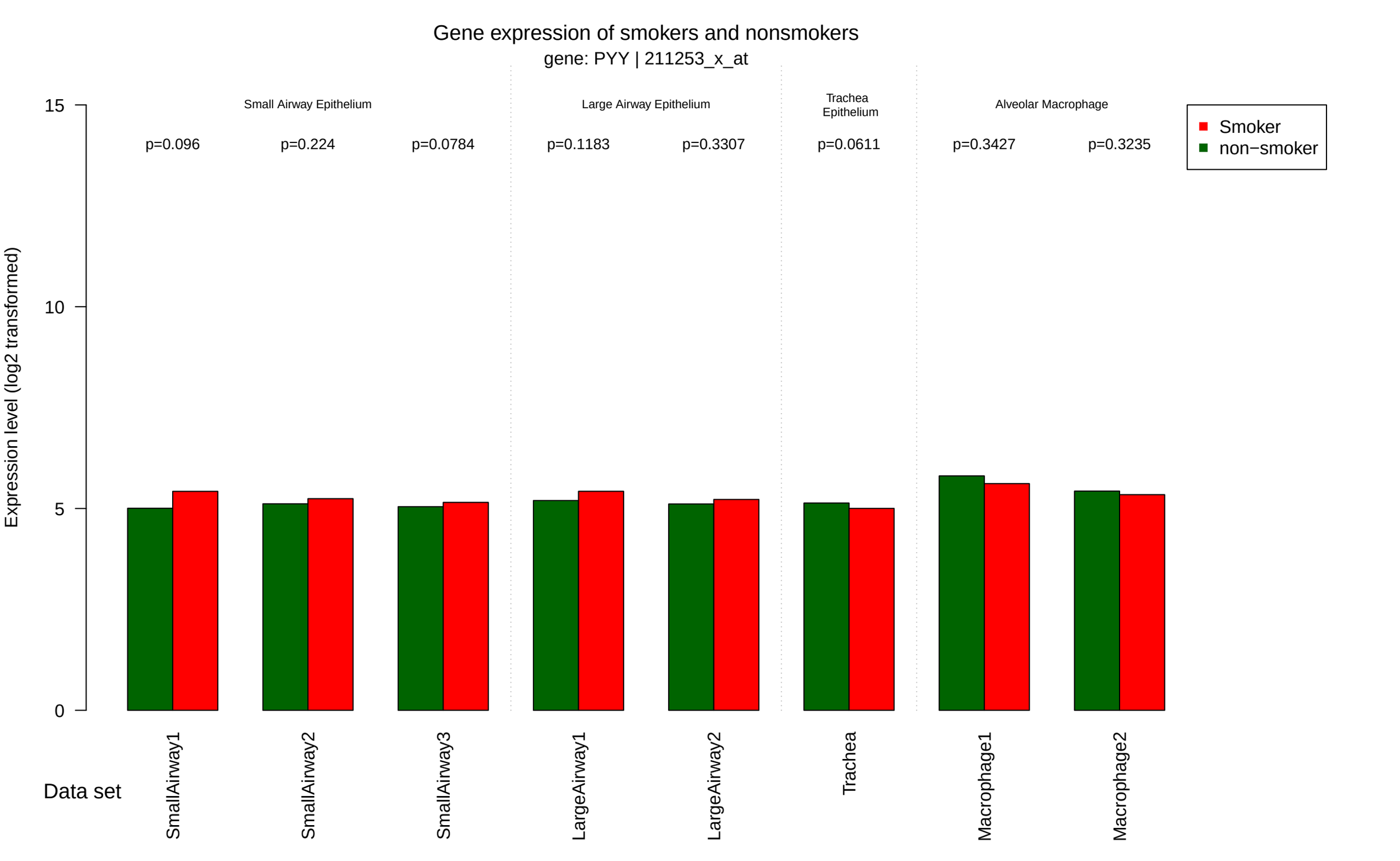
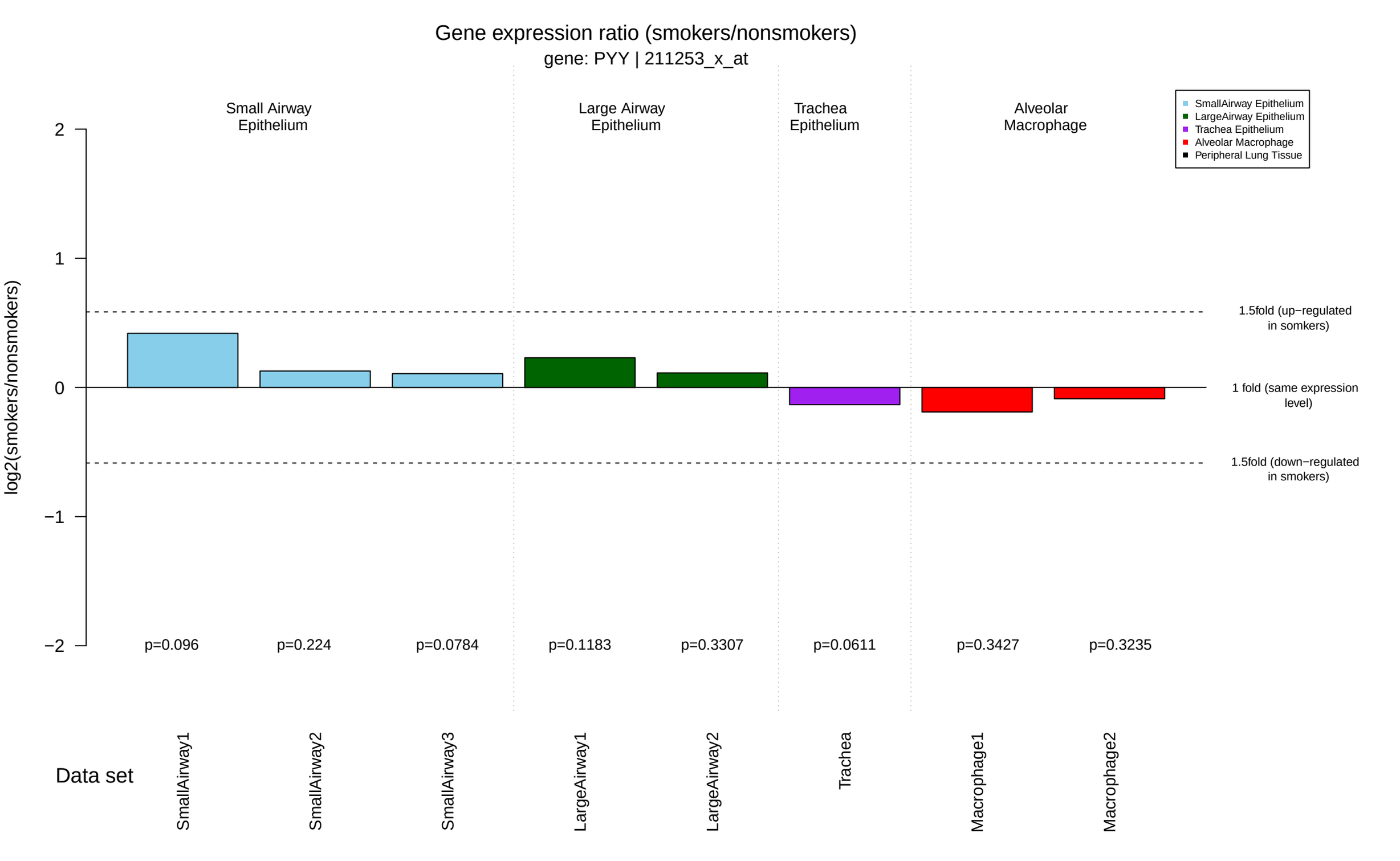
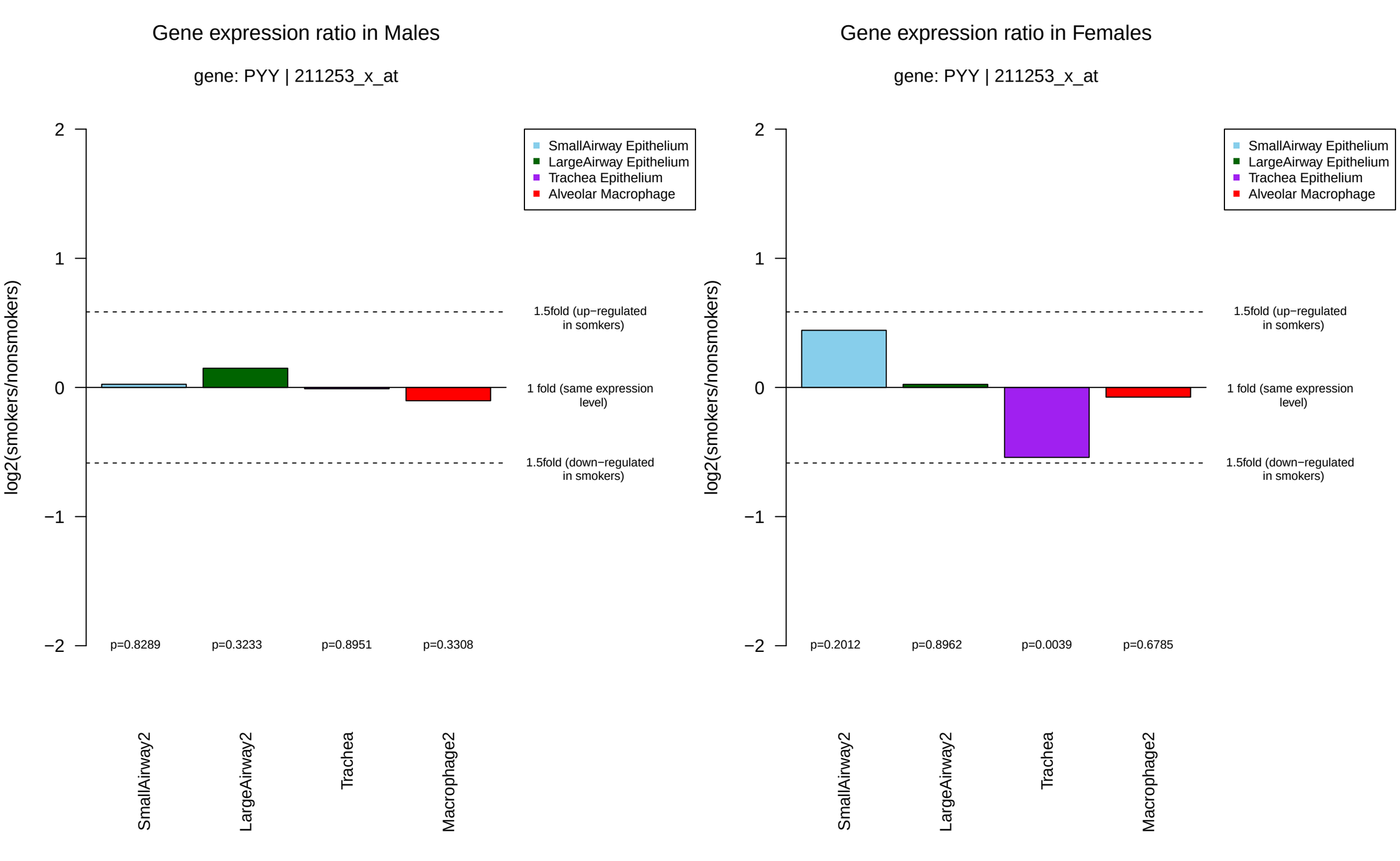
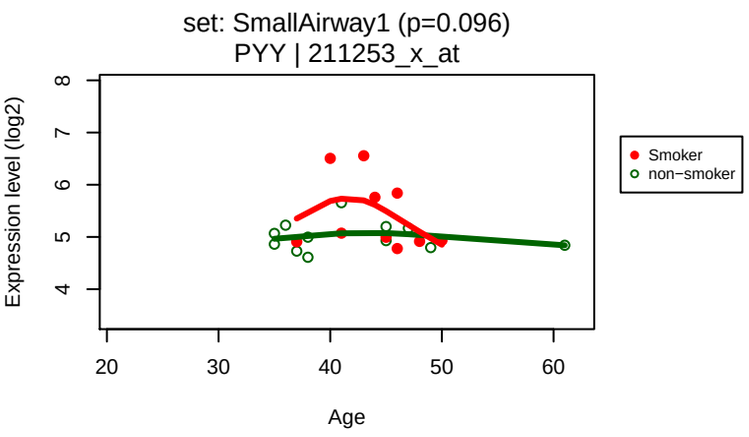
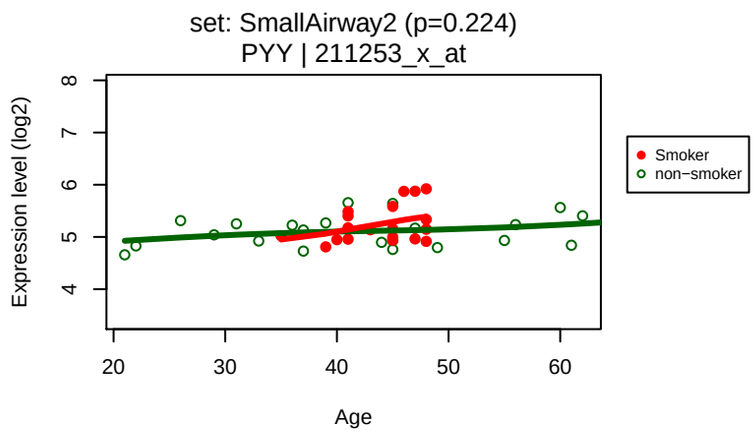
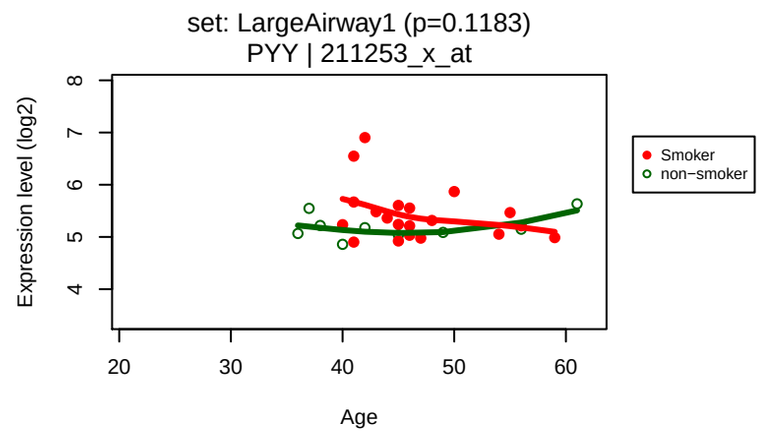

Gene expression of smokers and nonsmokers
gene: PYY | 211253_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.096
p=0.224
p=0.0784
p=0.1183
p=0.3307
p=0.0611
p=0.3427
p=0.3235
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

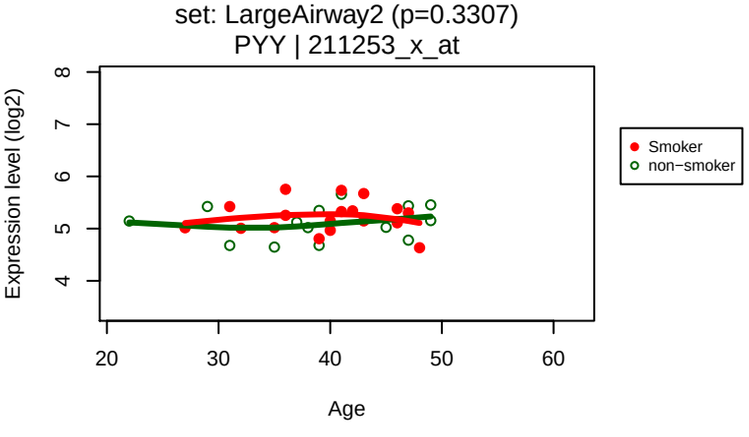

Gene expression ratio (smokers/nonsmokers)
gene: PYY | 211253_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.096
p=0.224
p=0.0784
p=0.1183
p=0.3307
p=0.0611
p=0.3427
p=0.3235
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

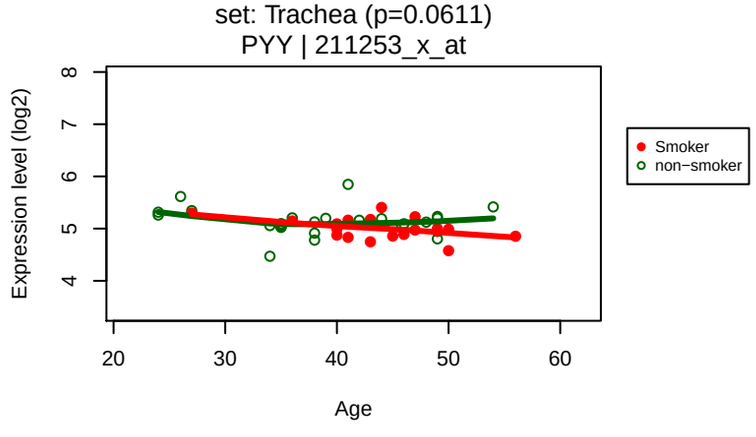

Gene expression ratio in Males
gene: PYY | 211253_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8289
p=0.3233
p=0.8951
p=0.3308
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PYY | 211253_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2012
p=0.8962
p=0.0039
p=0.6785
SmallAirway2
LargeAirway2
Trachea
Macrophage2

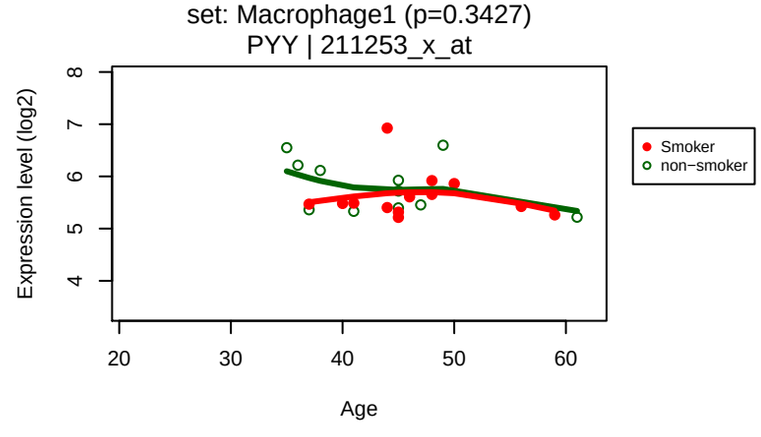

set: SmallAirway1 (p=0.096)
PYY | 211253_x_at
Expression level (log2)
Age
Smoker
non−smoker

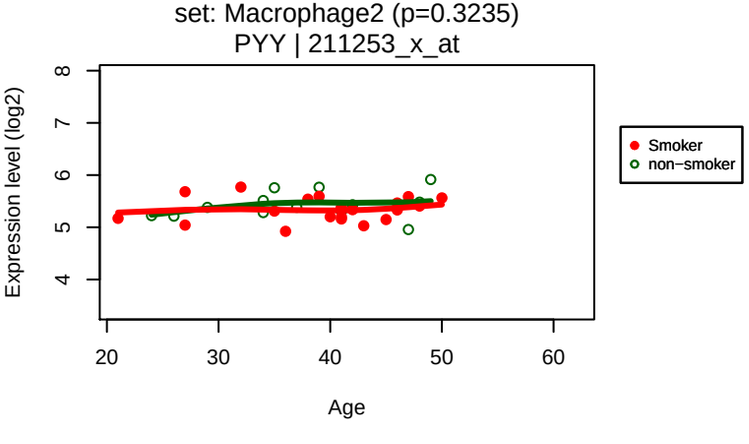

set: SmallAirway2 (p=0.224)
PYY | 211253_x_at
Expression level (log2)
Age
Smoker
non−smoker

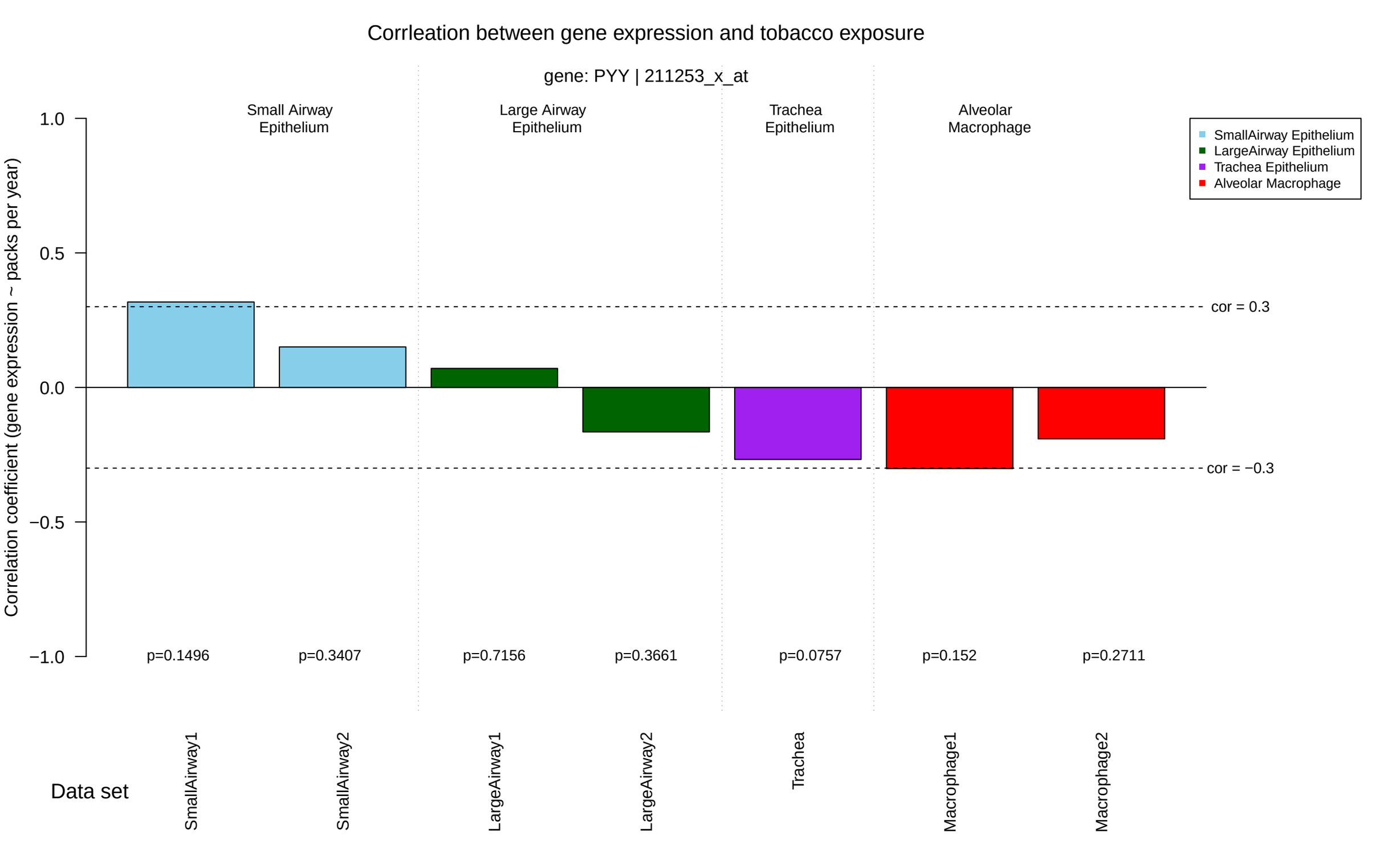

set: LargeAirway1 (p=0.1183)
PYY | 211253_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3307)
PYY | 211253_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0611)
PYY | 211253_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3427)
PYY | 211253_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3235)
PYY | 211253_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PYY | 211253_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1496
p=0.3407
p=0.7156
p=0.3661
p=0.0757
p=0.152
p=0.2711
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2